HW 5

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Let $S = \sum_{i=1}^{n} \epsilon_i A_i$. Assume $v > 0$. Since ϵ_i is Rademacher, $P(\epsilon_i = 1) = P(\epsilon_i = -1) = \frac{1}{2}$. Therefore, for any $\lambda \in \mathbb{R}$, $E[e^{\lambda \epsilon_i A_i}] = \frac{1}{2}e^{\lambda A_i} + \frac{1}{2}e^{-\lambda A_i} = \cosh(\lambda A_i)$. Note that for any scalar $x \in \mathbb{R}$, $\cosh(x) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$. We can rewrite $(2k)! = \prod_{i=1}^{k}(2j-1)(2j) \geq 2^k k!$. Therefore, $cosh(x) \leq \exp(\frac{x^2}{2})$. Because A_i is symmetric, it admits an eigendecompisition $A_i = U_i a_i U_i^\top$ where U_i is orthogonal and a_i is diagonal. Applying the scalar bound to each eigenvalue then gets us $cosh(\lambda A_i) \preceq \exp(\frac{\lambda^2 A_i^2}{2})$. Therefore, $E[e^{\lambda \epsilon_i A_i}] \preceq \exp(\frac{\lambda^2 A_i^2}{2})$. Both sides are positive definite, so the log also applies: $\log E[e^{\lambda \epsilon_i A_i}] \preceq \frac{\lambda^2 A_i^2}{2}$.

We now apply Lieb's inequality, that for symmetric random matrices X_i, $E[\operatorname{Tr}\exp(\sum_i X_i)] \leq \operatorname{Tr}\exp(\sum_i \log E[e^{X_i}])$. Let $X_i = \lambda \epsilon_i A_i$. Then, $E[\operatorname{Tr} e^{\lambda S}] \leq \operatorname{Tr}\exp(\sum_i \log E[e^{\lambda \epsilon_i A_i}]) \leq \operatorname{Tr}\exp(\frac{\lambda^2}{2}\sum_i A_i^2)$. Let $B = \sum_i A_i^2$. Since each $A_i^2 \succeq 0$, $B \succeq 0$, and $\lambda_{\max}(B) = \|B\|_{\text{op}} = v$. Thus, every eigenvalue of B is at most v, and $\operatorname{Tr}\exp(\frac{\lambda^2}{2}B) = \sum_{i=1}^{d} \exp(\frac{\lambda^2}{2}\lambda_i(B)) \leq d\exp(\frac{\lambda^2 v}{2})$. Therefore, $E[\operatorname{Tr} e^{\lambda S}] \leq d\exp(\frac{\lambda^2 v}{2})$.

For any $\lambda > 0$, Markov's inequality gives us $P(\lambda_{\max}(S) \geq t) = P(e^{\lambda \lambda_{\max}(S)} \geq e^{\lambda t}) \leq e^{-\lambda t} E[e^{\lambda \lambda_{\max}(S)}]$. Because $e^{\lambda S}$ is positive definite, $e^{\lambda \lambda_{\max}(S)} = \lambda_{\max}(e^{\lambda S}) \leq \operatorname{Tr} e^{\lambda S}$. We can combine this now with the previous inequality to get $P(\lambda_{\max}(S) \geq t) \leq e^{-\lambda t} E[\operatorname{Tr} e^{\lambda S}] \leq d e^{-\lambda t}\exp(\frac{\lambda^2 v}{2}) = d\exp(-\lambda t + \frac{\lambda^2 v}{2})$. Minimize the exponent by differentiating with respect to λ and setting to zero:

$$\frac{d}{d\lambda}(-\lambda t + \frac{\lambda^2 v}{2}) = -t + \lambda v = 0$$

$$\lambda^* = \frac{t}{v}$$

$$-\lambda^* t + \frac{(\lambda^*)^2 v}{2} = -\frac{t^2}{v} + \frac{t^2}{2v} = -\frac{t^2}{2v}$$

Thus, $P(\lambda_{\max}(S) \geq t) \leq d\exp(-\frac{t^2}{2v})$. By symmetry (since ϵ_i is Rademacher, ϵ_i and $-\epsilon_i$ have the same distribution), we also have $P(-\lambda_{\min}(S) \geq t) = P(\lambda_{\max}(-S) \leq t) \leq d\exp(-\frac{t^2}{2v})$. Again, because S is symmetric, $\|S\|_{\text{op}} = \max(\lambda_{\max}(S), -\lambda_{\min}(S))$. Using the union bound:

$$\begin{aligned} P(\|S\|_{\text{op}} \geq t) &= P(\lambda_{\max}(S) \geq t \text{ or } -\lambda_{\min}(S) \geq t) \\ &\leq P(\lambda_{\max}(S) \geq t) + P(-\lambda_{\min}(S) \geq t) \\ &\leq 2d\exp(-\frac{t^2}{2v}) \end{aligned}$$

Finally, if $v = 0$, then $\sum_{i=1}^n A_i^2 = 0$, which implies $A_i = 0$ for all i. Therefore, $S = 0$, and $\|S\|_{\text{op}} = 0$ with probability 1. Therefore, $P\left[\|\sum_{i=1}^n \epsilon_i A_i\|_{\text{op}} \geq t\right] \leq 2d\exp(-\frac{t^2}{2\|\sum_{i=1}^n A_i^2\|_{\text{op}}})$ for all $t \geq 0$.